

Problem 3: Sick Patient Example

Hidden Markov Model (HMM) — Filtering, Prediction & Viterbi

1. Given Data

States: Healthy (H), Sick (S) | **Initial probabilities:** $P(H) = 0.8$, $P(S) = 0.2$

Transition Matrix A (From / To)

From \ To	Healthy (H)	Sick (S)
Healthy (H)	0.7	0.3
Sick (S)	0.4	0.6

Emission (Observation) Probabilities B

State	Fever (F)	No Fever (N)
Healthy (H)	0.1	0.9
Sick (S)	0.8	0.2

Observation sequence: Fever $\rightarrow$ Fever $\rightarrow$ No fever (i.e. $o_1 = F$, $o_2 = F$, $o_3 = N$)

2. Filtering Probabilities (Forward Algorithm)

Filtering computes the posterior probability of the hidden state at time t given all observations up to time t : $P(X_t | o_{1:t})$. We use the forward algorithm. Define $\alpha_t(i) = P(o_{1:t}, X_t = i)$.

Time $t = 1$ (observation = Fever)

$$\alpha_1(H) = P(H) \cdot P(F|H) = 0.8 \times 0.1 = \mathbf{0.08}$$

$$\alpha_1(S) = P(S) \cdot P(F|S) = 0.2 \times 0.8 = \mathbf{0.16}$$

$$\text{Normalize: } P(X_1=H | o_1) = 0.08 / 0.24 = \mathbf{0.3333} \quad | \quad P(X_1=S | o_1) = 0.16 / 0.24 = \mathbf{0.6667}$$

Time $t = 2$ (observation = Fever)

$$\alpha_2(H) = [\alpha_1(H) \cdot 0.7 + \alpha_1(S) \cdot 0.4] \cdot 0.1 = (0.056 + 0.064) \times 0.1 = \mathbf{0.012}$$

$$\alpha_2(S) = [\alpha_1(H) \cdot 0.3 + \alpha_1(S) \cdot 0.6] \cdot 0.8 = (0.024 + 0.096) \times 0.8 = \mathbf{0.096}$$

$$\text{Normalize: } P(X_2=H | o_{1:2}) = 0.012 / 0.108 = \mathbf{0.1111} \quad | \quad P(X_2=S | o_{1:2}) = 0.096 / 0.108 = \mathbf{0.8889}$$

Time $t = 3$ (observation = No fever)

$$\alpha_3(H) = [\alpha_2(H) \cdot 0.7 + \alpha_2(S) \cdot 0.4] \cdot 0.9 = (0.0084 + 0.0384) \times 0.9 = \mathbf{0.04212}$$

$$\alpha_3(S) = [\alpha_2(H) \cdot 0.3 + \alpha_2(S) \cdot 0.6] \cdot 0.2 = (0.0036 + 0.0576) \times 0.2 = \mathbf{0.01224}$$

$$\text{Normalize: } P(X_3=H | o_{1:3}) = 0.04212 / 0.05436 \approx \mathbf{0.7749} \quad | \quad P(X_3=S | o_{1:3}) = 0.01224 / 0.05436 \approx \mathbf{0.2251}$$

Filtering Probabilities Summary

t	$P(H o_{1:t})$	$P(S o_{1:t})$
1	0.3333	0.6667
2	0.1111	0.8889
3	0.7749	0.2251

3. Predict the Patient's Health State

Prediction of the next state X_4 given all observations so far ($o_{1:3}$):

$$P(X_4=H \mid o_{1:3}) = P(X_3=H) \cdot 0.7 + P(X_3=S) \cdot 0.4 = 0.7749 \times 0.7 + 0.2251 \times 0.4 = \mathbf{0.6324}$$

$$P(X_4=S \mid o_{1:3}) = P(X_3=H) \cdot 0.3 + P(X_3=S) \cdot 0.6 = 0.7749 \times 0.3 + 0.2251 \times 0.6 = \mathbf{0.3676}$$

Predicted next state: Healthy (probability ≈ 0.632)

4. Most Likely Hidden-State Sequence (Viterbi Algorithm)

We find the single most probable path X_1, X_2, X_3 that generated the observations. Define $V_t(i) = \max P(x_{1:t-1}, X_t=i, o_{1:t})$.

Initialization (t = 1)

$$V_1(H) = 0.8 \times 0.1 = \mathbf{0.08} \quad V_1(S) = 0.2 \times 0.8 = \mathbf{0.16}$$

Recursion (t = 2)

$$V_2(H) = \max(V_1(H) \cdot 0.7, V_1(S) \cdot 0.4) \times 0.1 = \max(0.056, 0.064) \times 0.1 = \mathbf{0.0064} \text{ (from S)}$$

$$V_2(S) = \max(V_1(H) \cdot 0.3, V_1(S) \cdot 0.6) \times 0.8 = \max(0.024, 0.096) \times 0.8 = \mathbf{0.0768} \text{ (from S)}$$

Recursion (t = 3)

$$V_3(H) = \max(V_2(H) \cdot 0.7, V_2(S) \cdot 0.4) \times 0.9 = \max(0.00448, 0.03072) \times 0.9 = \mathbf{0.027648} \text{ (from S)}$$

$$V_3(S) = \max(V_2(H) \cdot 0.3, V_2(S) \cdot 0.6) \times 0.2 = \max(0.00192, 0.04608) \times 0.2 = \mathbf{0.009216} \text{ (from S)}$$

Termination & Backtracking

$$\max(V_3(H), V_3(S)) = 0.027648 \rightarrow \text{end state} = \mathbf{H}$$

$$\text{Backtrack: } X_3 = H \text{ (from S)} \rightarrow X_2 = S \text{ (from S)} \rightarrow X_1 = S$$

Most likely hidden-state sequence: S $\rightarrow$ S $\rightarrow$ H (Sick $\rightarrow$ Sick $\rightarrow$ Healthy)

Final Answers Summary

#	Question	Answer
1	Filtering probabilities	t=1: H=0.333, S=0.667 t=2: H=0.111, S=0.889 t=3: H=0.775, S=0.225
2	Predict next health state	Healthy (P ≈ 0.632)
3	Most likely state sequence	S $\rightarrow$ S $\rightarrow$ H

Subject: Artificial Intelligence 2 | Topic: Hidden Markov Models